ARAZVERA TECHNOLOGIES • ENGINEERING PORTFOLIO • NOV 2026

LEVEL 4
Enterprise Resilience & Security Engineering

Evidence-first architecture, failure engineering and secure multi-service reliability

Position: A real, synthetic, single-host engineering laboratory. Not a customer production deployment, uptime warranty or independent security certification.

Engineering proposition

Complex backends fail where APIs, identity, PostgreSQL transactions, workers, monitoring and releases intersect. ARAZVERA Level 4 provides scoped investigation, reproducible failures, safe interventions and measurable evidence.

Specialist services

Area	Typical scope
Architecture & Reliability Audit	Dependency mapping, failure pathways, measured risk gaps and remediation acceptance plans
Multi-Tenant Security Hardening	PostgreSQL FORCE RLS, restricted roles, RS256/JWKS validation and negative authorization tests
Recovery Engineering	Transactional outbox, lease fencing, bounded retries, idempotency, dead letters and Worker recovery
Safe Release & Rollback	200-request synthetic candidate observation, fail-closed HOLD, stable/candidate traffic routing and rollback
Incident & SLO Readiness	Real health probes, error/latency metrics, durable incident history, explicit sample limits
Backend Reliability Remediation	Authorized FastAPI/Python/PostgreSQL/Docker changes with repeatable engineering tests

Verified proof — local synthetic environment

Test group	Measured result
Private GitHub Actions	**4 / 4 jobs successful** at commit a1b6e137; run 42763566633

Local Docker Pytest suite	**89 passed, 0 failed**; two non-blocking cache warnings
Python unit tests	**67 passed**
Real Selenium/Chrome interface checks	**13 / 13 passed**, including Canary buttons
Real synthetic Canary HTTP acceptance	**3 / 3 passed**; healthy candidate, failing candidate HOLD and rollback
Local PostgreSQL 16 logical restore	**8 / 8 passed** in a second disposable database container
Scoped Worker restart	**7 / 7 passed** with durable receipt after recovery

Counts across suites overlap and must not be added together. Baseline and current private CI evidence requires authorized repository access.

Architecture snapshot

The system contains a FastAPI Control Plane, isolated PostgreSQL 16, independent Worker, periodic observer, ephemeral laboratory RS256 identity, fixed synthetic downstream, and two separate Stable/Candidate services. Synthetic components operate within an isolated Level 4 Docker project.

A Canary evaluation sends **200 actual HTTP requests** to the candidate and stores measured status/latency evidence. A healthy candidate may activate the lab-only route; a faulty candidate triggers HOLD and atomically restores Stable. This neither deploys an image nor changes a customer service.

Engagement design

1. **Define:** client-authorized infrastructure boundary, failure cases, acceptance criteria and exclusions.
2. **Reproduce:** baseline failures and authorization negatives using safe tests.
3. **Engineer:** scoped backend, database, identity or release changes.
4. **Verify:** repeat measurable regressions and durability/failure tests.
5. **Handover:** code review, audit evidence, runbook, release/rollback assumptions.

Commercial scope is proposal-specific: fixed-price milestones or time-and-materials under written acceptance. Client-owned cloud, hosting, licenses and credentials require separate approval. No price or contract outcome is guaranteed.

Explicit limits and exclusions

- No verified customer production deployment, customer IAM/SSO integration, Enterprise security certification or formal SLA.
- Single-host logical restore does not establish independent-site disaster recovery, customer RPO/RTO or multi-host failover.
- Staging canary/rollback routes are strictly synthetic; no production application version is changed.
- SHA-256 is a content checksum, **not** a signed attestation identifying an operator.
- Basic security/secret scans do not replace an independent audit, SBOM or pinned supply-chain provenance.
- No keys, private implementation source, customer data or third-party production systems are disclosed.

ARAZVERA TECHNOLOGIES

Company GitHub

Engineering Portfolio

Specialist capability statement, not a warranty or claim of historical Enterprise projects.